



PCB LAYOUT GUIDE

JMS581-Series USB 3.2 Gen2x1 to SATA 6Gb/s, PCIe G3x2 & SD Express Bridge Controller

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Revision History

Revision number	Effective date	Description of revision		Author
		Reference	Description of change	
1.00	07/10/2020	--	Initial release.	Jason Hsiao
1.01	01/12/2021	1.00	1. Modify PCIe3.0 、SD7.0 impedance description 2. Modify power layout description	Jason Hsiao
1.02	02/21/2024	--	1. Modify SD7.0 impedance description 2. Add CFexpress layout rule	Mika Cheng

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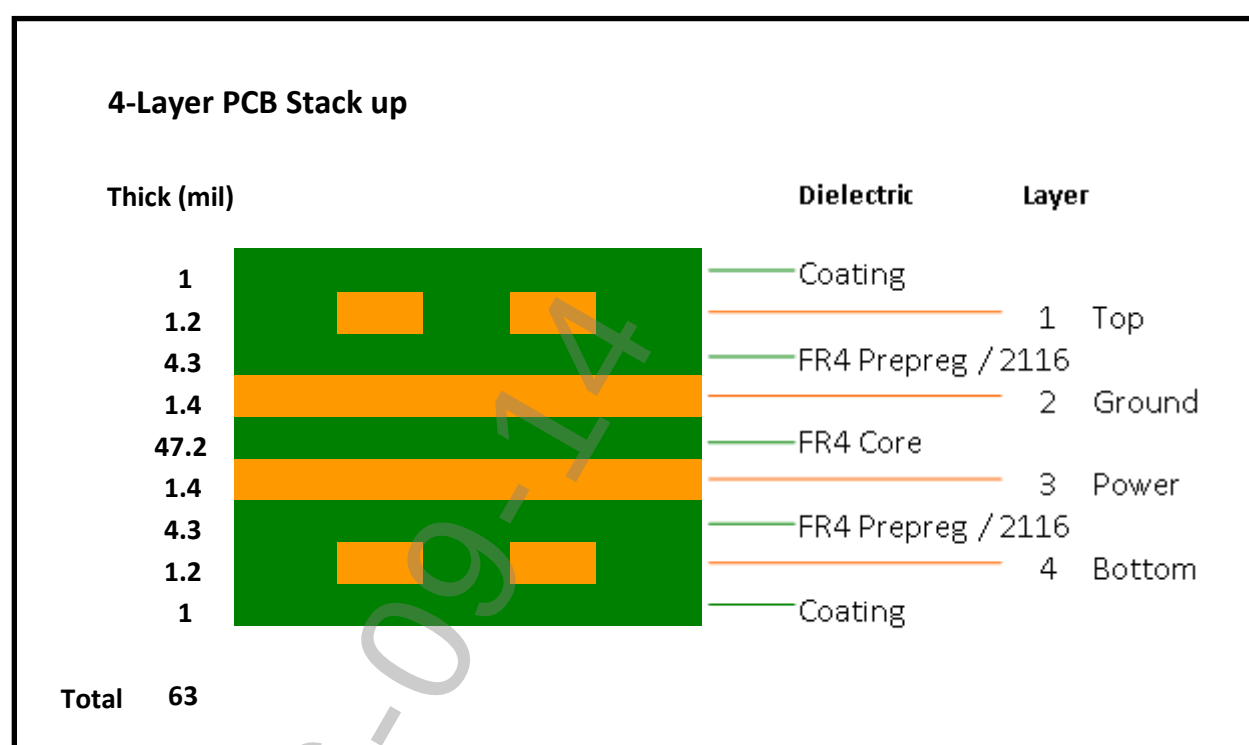
1 Overview

This Layout Guide can be applied to JMS581 series.

1.1 Description

This layout guide includes USB3.2, PCIe3.0, SATA3.0, USB2.0, SD7.0, Power plane, Crystal and Switching Regulator.

1.2 PCB stack up



2 USB3.2 layout guide

2.1 Relative net name & pairs

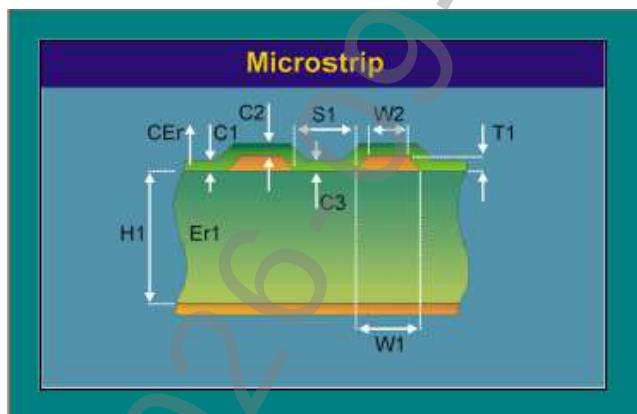
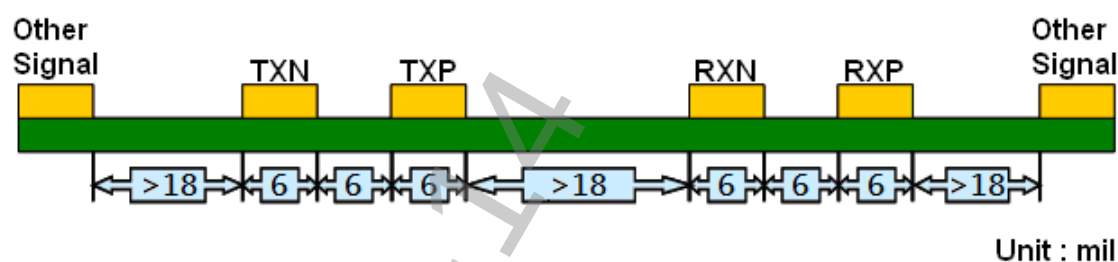
USB3.2 have 4 differential signal pair , detailed information is as follows:

Net name	Routing layer	Reference layer
U3_TXP1, U3_TXN1, U3_RXN1, U3_RXP1, U3_TXP2, U3_TXN2, U3_RXN2, U3_RXP2	1st layer	2nd layer (GND)

2.2 Net spacing & trace length rule

- USB3.2 Impedance : 90 Ω +/-10 Ω
- USB3.2 Max Trace length : 6.5 inch (16.51cm)
- USB3.2 Trace : Trace width / trace separation / pair separation = 6 / 6 / 18.0 mil

Target differential impedance: 90 Ω

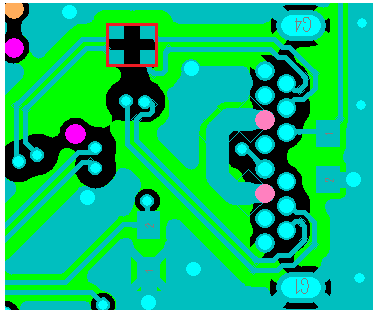


		Tolerance	Minimum	Maximum	
H1	4.3000	+/- 0.0000	4.3000	4.3000	Calculate
Er1	4.2000	+/- 0.0000	4.2000	4.2000	Calculate
W1	6.0000	+/- 0.0000	6.0000	6.0000	
W2	6.0000	+/- 0.0000	6.0000	6.0000	Calculate
S1	6.0000	+/- 0.0000	6.0000	6.0000	Calculate
T1	1.2000	+/- 0.0000	1.2000	1.2000	Calculate
C1	1.0000	+/- 0.0000	1.0000	1.0000	
C2	1.0000	+/- 0.0000	1.0000	1.0000	
C3	1.0000	+/- 0.0000	1.0000	1.0000	
CEr	3.4000	+/- 0.0000	3.4000	3.4000	
Zdiff	90.79		90.79	90.79	Calculate

- USB3.2 trace length mismatch < 5mil.
- Away from the Oscillator, Switching Regulator LC, Power components and Power trace.
- Route all SuperSpeed USB signal traces over continuous planes (VCC or GND), with no interruptions. Avoid crossing over anti-etch, commonly found with plane splits.
- Do not route SuperSpeed USB traces under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that use or duplicate clock signals.

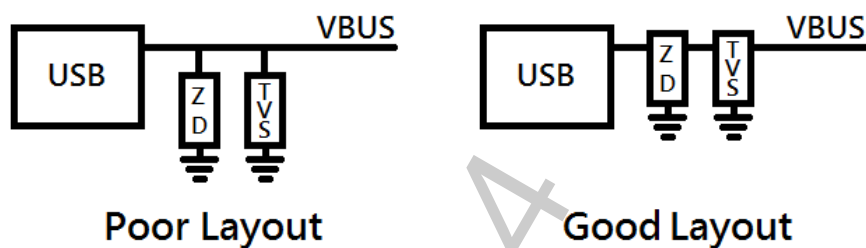
2.3 USB Type-C connector example

Layout example for USB connector



Note: Remove area under capacitors

Layout example for Zener Diode and TVS :



3 PCIe3.0 layout guide

3.1 Related net name & pairs

PCIE3.0 have 5 differential signal pair , detailed information is as follows:

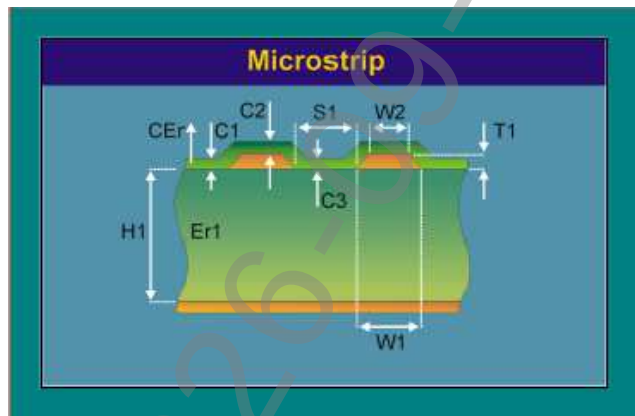
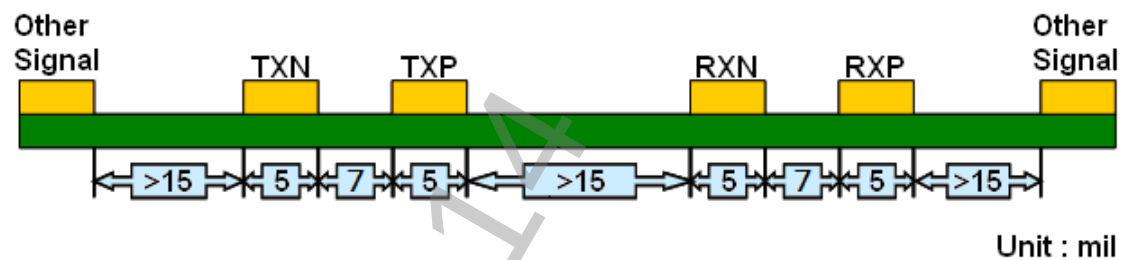
Net name	Routing layer	Reference layer
P0_TXP0, P0_TXN0, P0_RXN0, P0_RXP0, P0_TXP1, P0_TXN1, P0_RXN1, P0_RXP1, P0_CLKP, P0_CLKN,	1st layer	2nd layer (GND)

3.2 Net spacing & trace length rule

- PCIe3.0 Max Trace length to PCIe Host : 11 inch (27.94 cm)
- For PCIe Slot , PCIe3.0 Impedance : 100 Ω +/-10 Ω

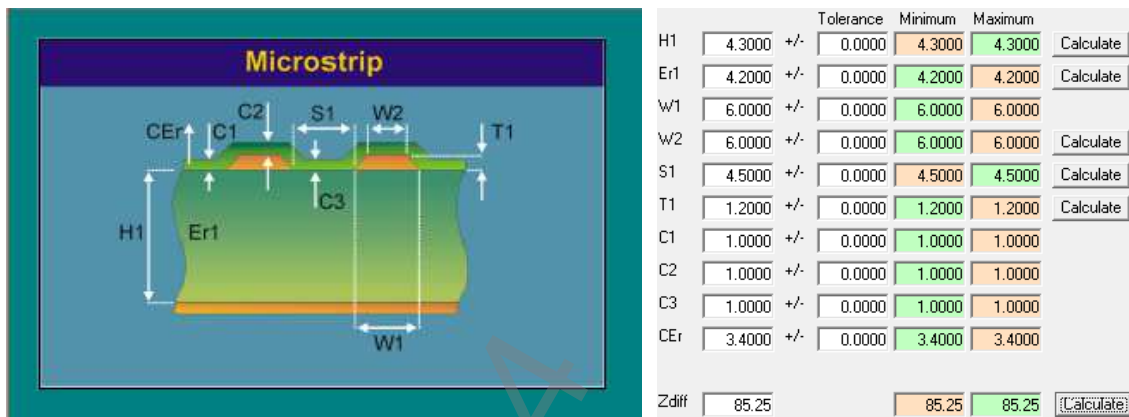
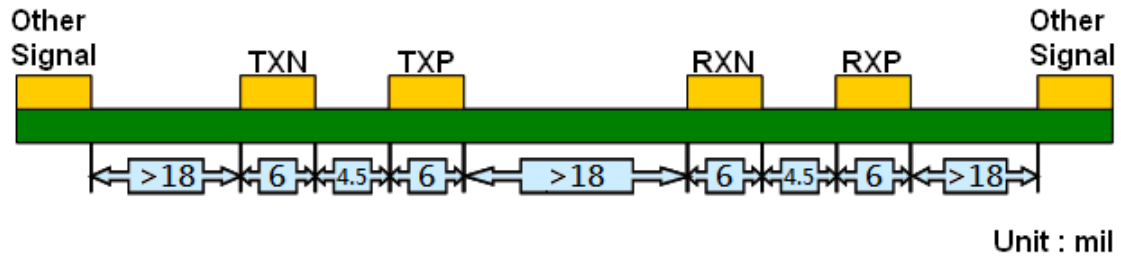
PCIE3.0 Trace : Trace Width / Trace Separation / Pair Separation = 5 / 7 / 15 mil]

PCIe slot target differential impedance: 100 Ω



		Tolerance	Minimum	Maximum	
H1	4.3000 +/-	0.0000	4.3000	4.3000	Calculate
Er1	4.2000 +/-	0.0000	4.2000	4.2000	Calculate
W1	5.0000 +/-	0.0000	5.0000	5.0000	
W2	5.0000 +/-	0.0000	5.0000	5.0000	Calculate
S1	7.0000 +/-	0.0000	7.0000	7.0000	Calculate
T1	1.2000 +/-	0.0000	1.2000	1.2000	Calculate
C1	1.0000 +/-	0.0000	1.0000	1.0000	
C2	1.0000 +/-	0.0000	1.0000	1.0000	
C3	1.0000 +/-	0.0000	1.0000	1.0000	
CEr	3.4000 +/-	0.0000	3.4000	3.4000	
Zdiff	100.74		100.74	100.74	Calculate

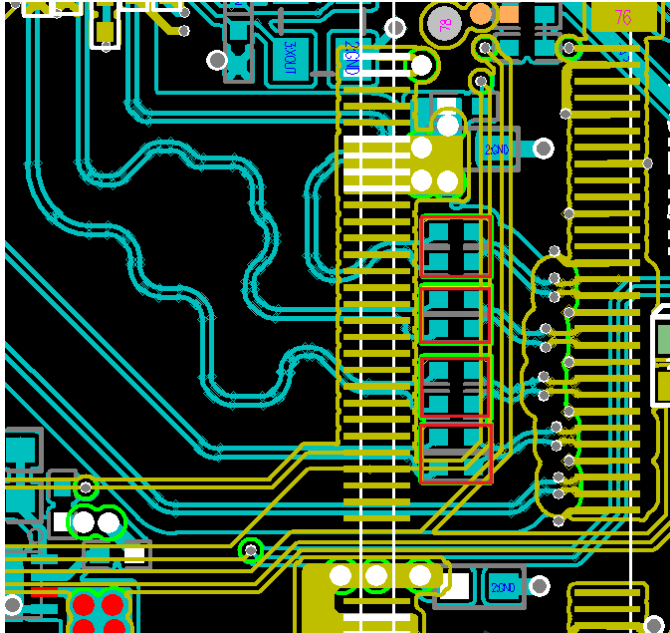
- For PCIe M.2 , PCIe3.0 Impedance : 85 Ω + /-10 Ω
 PCIe3.0 Trace : Trace Width / Trace Separation / Pair Separation = 6 / 4.5 / 18 mil]
 M.2 target differential impedance: 85 Ω



- P0_TXP0, P0_TXN0, P0_TXP1, P0_TXN1 trace length mismatch < 5mil.
- P0_RXP0, P0_RXN0, P0_RXP1, P0_RXN1 trace length mismatch < 5mil.
- P0_CLKP, P0_CLKN trace length mismatch < 5mil.
- Away from the Oscillator, Switching Regulator LC, Power components and Power trace.
- Route all PCIe signal traces over continuous planes (VCC or GND), with no interruptions. Avoid crossing over anti-etch, commonly found with plane splits.
- Do not route PCIe traces under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that use or duplicate clock signals.

3.3 PCIe M.2 connector example

Layout example for M.2 connector



Note: Remove area under capacitors

4 SATA3.0 layout guide

4.1 Related net name & pairs

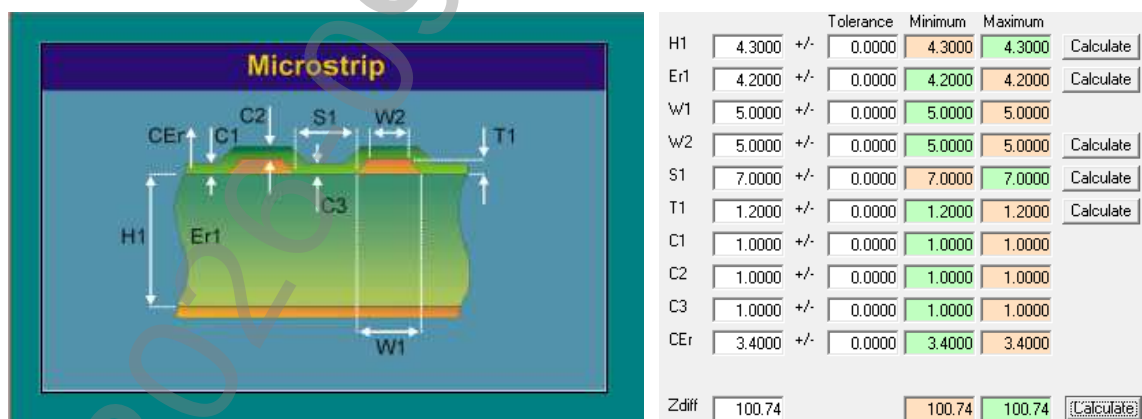
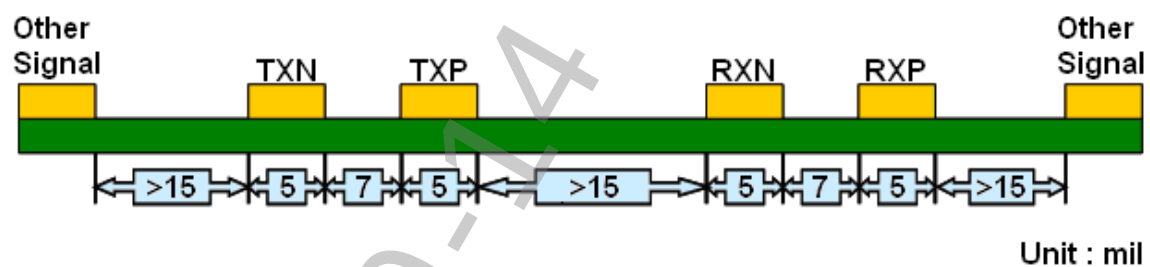
SATA3.0 have 2 differential signal pair , detailed information is as follows:

Net name	Routing layer	Reference layer
SA_TXP, SA_TXN, SA_RXN, SA_RXP	1st layer	2nd layer (GND)

4.2 Net spacing & trace length rule

- SATA Impedance : $100\Omega \pm 10\Omega$
- SATA Max Trace length to SATA Host : 10 inch (25.4 cm)
- SATA Trace : Trace Width / Trace Separation / Pair Separation = 5 / 7 / 15 mil]

Target differential impedance: 100Ω

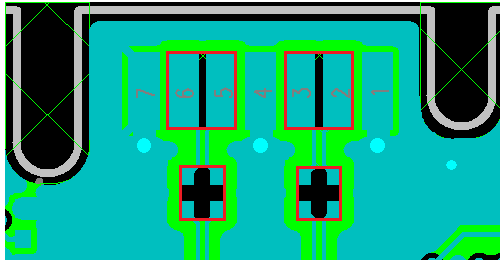


- SATA3.0 trace length should be between 0.7inch (1.8cm) and 10inch (25.4cm), and trace mismatch < 5mil.
- Away from the Oscillator, Switching Regulator LC, Power components and Power trace.
- Route all SATA signal traces over continuous planes (VCC or GND), with no interruptions. Avoid crossing over anti-etch, commonly found with plane splits.

- Do not route SATA traces under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that use or duplicate clock signals.

4.3 SATA connector example

Layout example for SATA connector



Note: Remove area under capacitors and signal pins

5 USB2.0 layout guide

5.1 Related net name & pairs

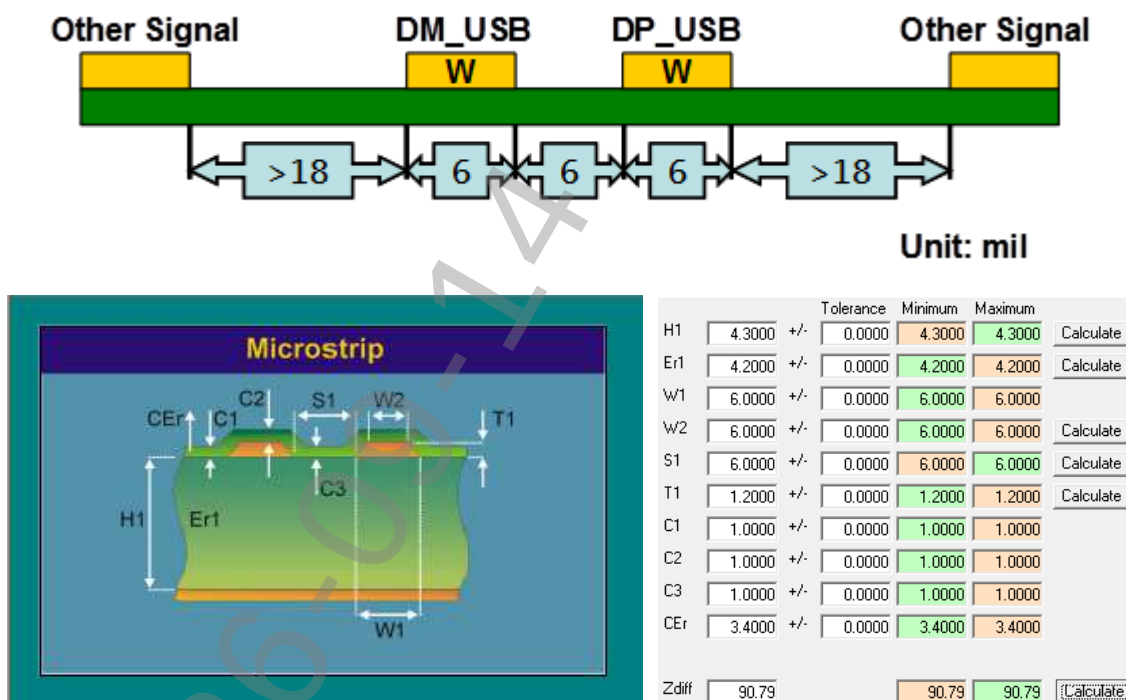
USB2.0 have a differential signal pair , detailed information is as follows:

Net name	Routing layer	Reference layer
U_DP, U_DM	1st layer	2nd layer (GND)

5.2 Net spacing & trace length rule

- USB2.0 Impedance : 90 Ω +/-10 Ω
- USB2.0 Max Trace length : 6.5 inch (16.51 cm)
- USB Trace : Trace width / trace separation / pair separation = 6 / 6 / 18 mil

Target differential impedance: 90 Ω



- USB2.0 trace length mismatch < 5mil.
- Away from the Oscillator, Switching Regulator LC, Power components and Power trace.
- Route all high-speed USB signal traces over continuous planes (VCC or GND), with no interruptions. Avoid crossing over anti-etch, commonly found with plane splits.
- Do not route high-speed USB traces under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that use or duplicate clock signals.
- The D+ and D- DC Resistance is the series combination of any resistance in switches, MUXes, and the USB PHY. The DC Resistance of the D+ and D- in USB Type-C Device shall be equal or less than the maximum

value specified in table below :

	Maximum
USB Type-C Device	19Ω
USB Type-C Captive Device	25Ω

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6 SD7.0 layout guide

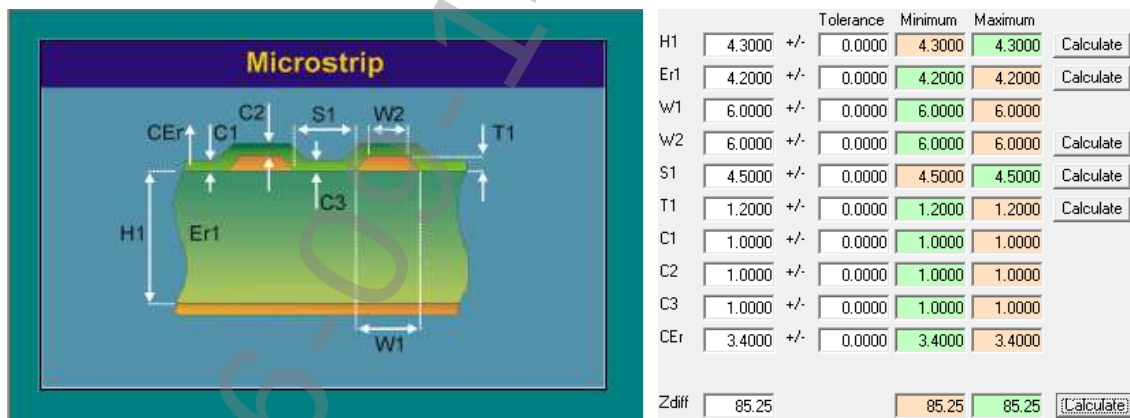
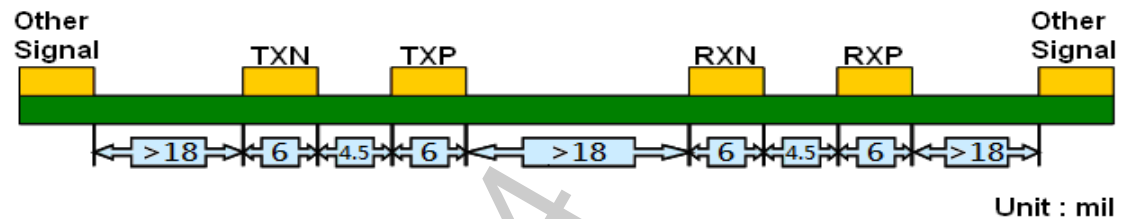
6.1 Related net name & pairs

SD7.0 have 3 differential signal pairs , detailed information is as follows:

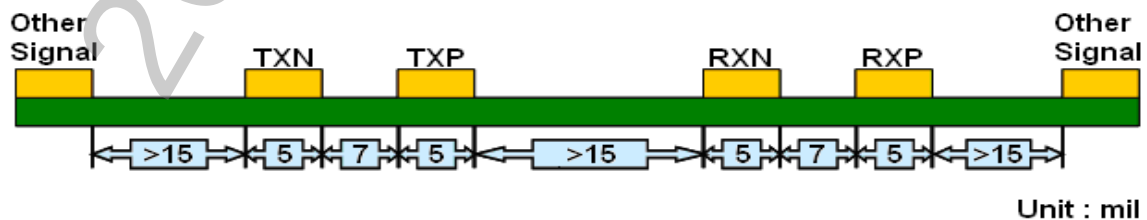
Net name	Routing layer	Reference layer
SD_D0, SD_D1, SD_D2, SD_D3, SD_CLK, SD_CMD, P1_TXP, P1_TXN, P1_RXP, P1_RXN	1st layer	2nd layer (GND)

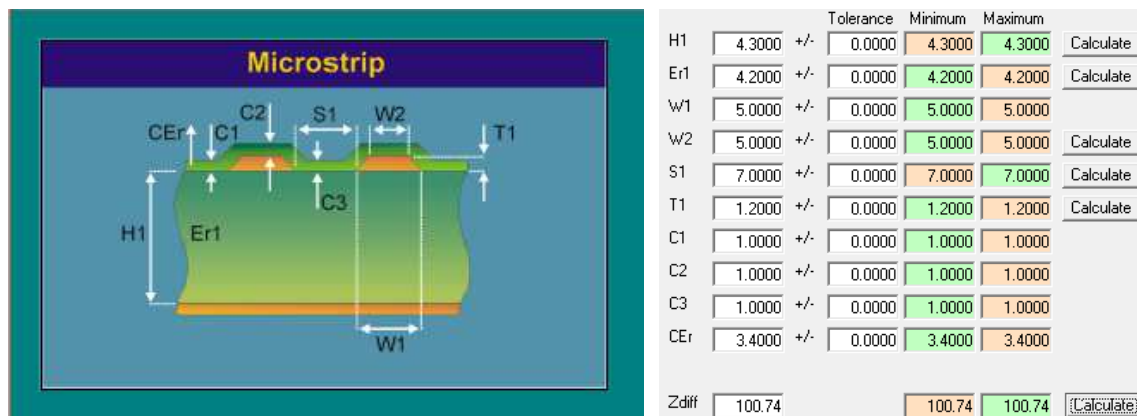
6.2 Net spacing & trace length rule

- SD7.0 Single End Max Trace Length : 2.5 inch (6.35 cm)
- SD7.0 Data Differential Impedance : $85 \Omega \pm 10 \Omega$ (for P1_TXP, P1_TXN, P1_RXP, P1_RXN)
- SD7.0 Data PCIe Trace : Trace width / trace separation / pair separation = 6 / 4.5 / 18 mil



- SD7.0 Clock Differential Impedance : $100 \Omega \pm 10 \Omega$ (for SD_D0, SD_D1)
- SD7.0 Differential Max Trace Length : 2.5 inch (6.35 cm)
- SD7.0 PCIe Trace : Trace width / trace separation / pair separation = 5 / 7 / 18 mil

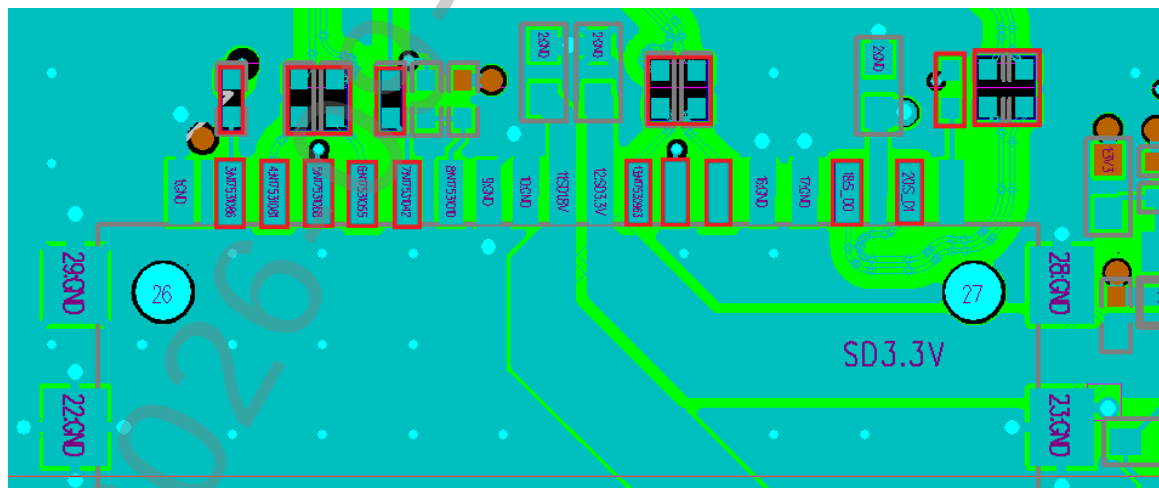




- SD7.0 PCIe trace length mismatch < 5mil.
- SD_CLK, SD_CMD, SD_D0, SD_D1, SD_D2, SD_D3 trace length mismatch < 10mil.
- Away from the Oscillator, Switching Regulator LC, Power components and Power trace.
- Route all high-speed SD7.0 signal traces over continuous planes (VCC or GND), with no interruptions. Avoid crossing over anti-etch, commonly found with plane splits.
- Do not route high-speed SD7.0 traces under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that use or duplicate clock signals.

6.3 SD7.0 connector example

Layout example for SD7.0 connector



Note: Remove area under capacitors and signal pins

7 CFexpress layout guide

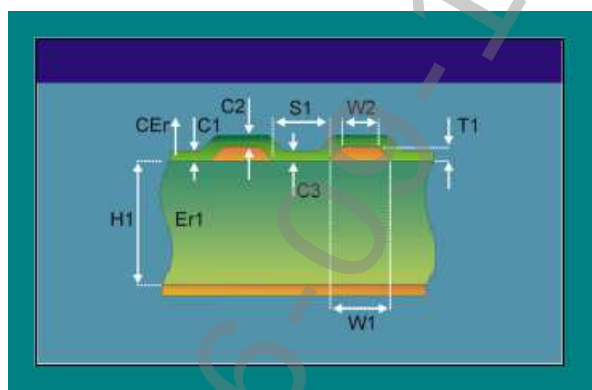
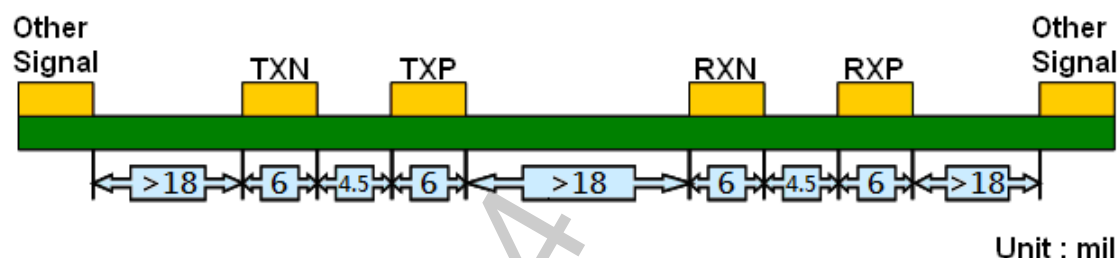
7.1 Related net name & pairs

CFexpress has 5 pairs of differential signals, detailed information is as follows:

Net name	Routing layer	Reference layer
REFCLK+, REFCLK-, PCIe TX0+, PCIe TX0-, PCIe TX1+, PCIe TX1-, PCIe RX0+, PCIe RX0-, PCIe RX1+, PCIe RX1-	1st layer	2nd layer (GND)

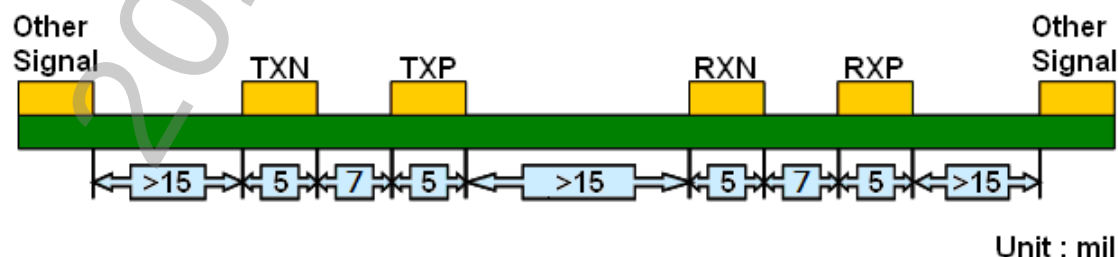
7.2 Net spacing & trace length rule

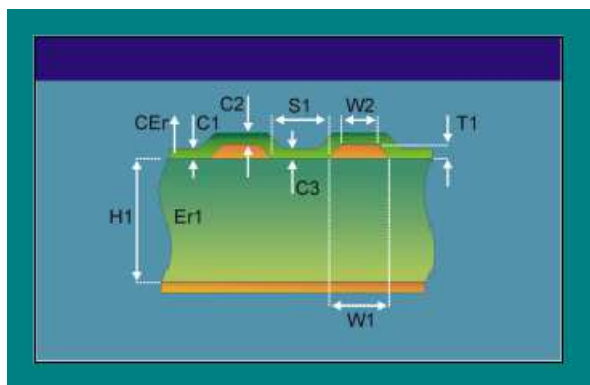
- CFexpress Data trace impedance: $85 \Omega \pm 15 \Omega$
- CFexpress Data trace : Trace Width / Trace Separation / Pair Separation = 6 / 4.5 / 18 mil.



		Tolerance	Minimum	Maximum	
H1	4.3000	+/- 0.0000	4.3000	4.3000	Calculate
Er1	4.2000	+/- 0.0000	4.2000	4.2000	Calculate
W1	6.0000	+/- 0.0000	6.0000	6.0000	Calculate
W2	6.0000	+/- 0.0000	6.0000	6.0000	Calculate
S1	4.5000	+/- 0.0000	4.5000	4.5000	Calculate
T1	1.2000	+/- 0.0000	1.2000	1.2000	Calculate
C1	1.0000	+/- 0.0000	1.0000	1.0000	
C2	1.0000	+/- 0.0000	1.0000	1.0000	
C3	1.0000	+/- 0.0000	1.0000	1.0000	
CEr	3.4000	+/- 0.0000	3.4000	3.4000	
Zdiff	85.25		85.25	85.25	Calculate

- CFexpress Clock trace impedance: $100 \Omega \pm 10 \Omega$
- CFexpress Clock trace : Trace Width / Trace Separation / Pair Separation = 5 / 7 / 15 mil.





			Tolerance	Minimum	Maximum	
H1	4.3000	+/-	0.0000	4.3000	4.3000	Calculate
Er1	4.2000	+/-	0.0000	4.2000	4.2000	Calculate
W1	5.0000	+/-	0.0000	5.0000	5.0000	
W2	5.0000	+/-	0.0000	5.0000	5.0000	Calculate
S1	7.0000	+/-	0.0000	7.0000	7.0000	Calculate
T1	1.2000	+/-	0.0000	1.2000	1.2000	Calculate
C1	1.0000	+/-	0.0000	1.0000	1.0000	
C2	1.0000	+/-	0.0000	1.0000	1.0000	
C3	1.0000	+/-	0.0000	1.0000	1.0000	
CEr	3.4000	+/-	0.0000	3.4000	3.4000	
Zdiff	100.74			100.74	100.74	Calculate

- CFexpress Differential Impedance :
 - $85 \Omega \pm 15 \Omega$ (for PCIe TX0+, PCIe TX0-, PCIe TX1+, PCIe TX1-, PCIe RX0+, PCIe RX0-, PCIe RX1+, PCIe RX1-)
 - $100 \Omega \pm 10 \Omega$ (for REFCLK+, REFCLK-)
- CFexpress Differential Max Trace Length : 2.5 inch (6.35 cm)
- CFexpress trace length mismatch < 5mil.

8 Crystal layout guide

8.1 Related net name & pairs

The Oscillator/Crystal detailed information is as follows:

Net name	Routing layer	Reference layer
XIN, XOUT	1st layer	2nd layer (GND)

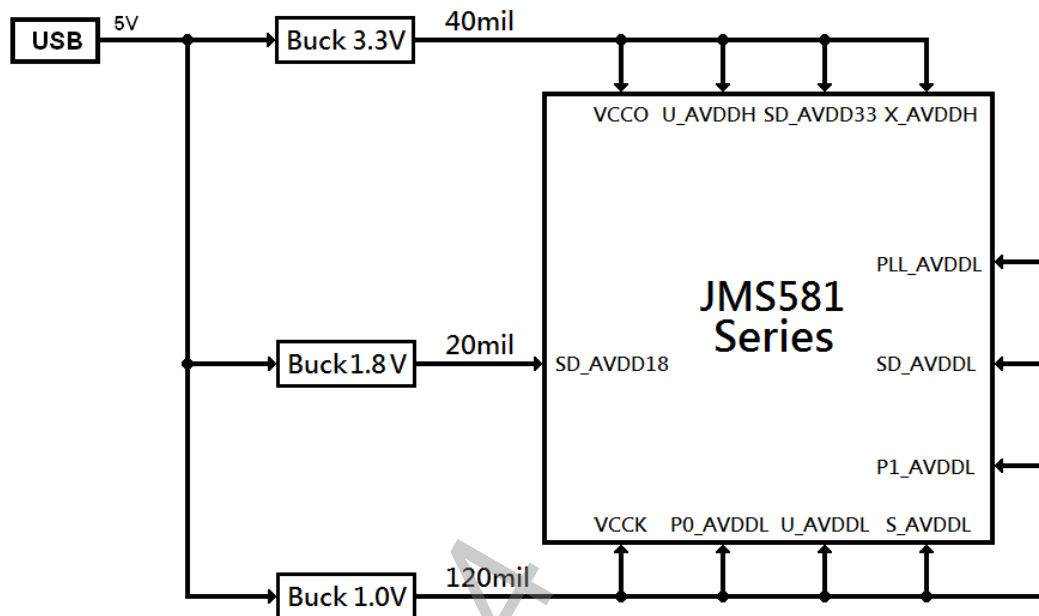
8.2 Layout rule

- The crystal unit should then be placed as close as possible to the XIN and XOUT pins to minimize etch lengths.
- Ensure that the ground plane under the IC and its components are of good quality.
- Avoid placing a separate ground under the oscillator and connecting it to the general ground through a single point.
- Avoid long connections to the crystal and to the load capacitor that create a large loop on the PCB.
- Use a short connection between the two crystal load capacitors and route the common connection to the IC ground reference.

9 Power layout guide

9.1 Related net name

The Power detailed information is as follows

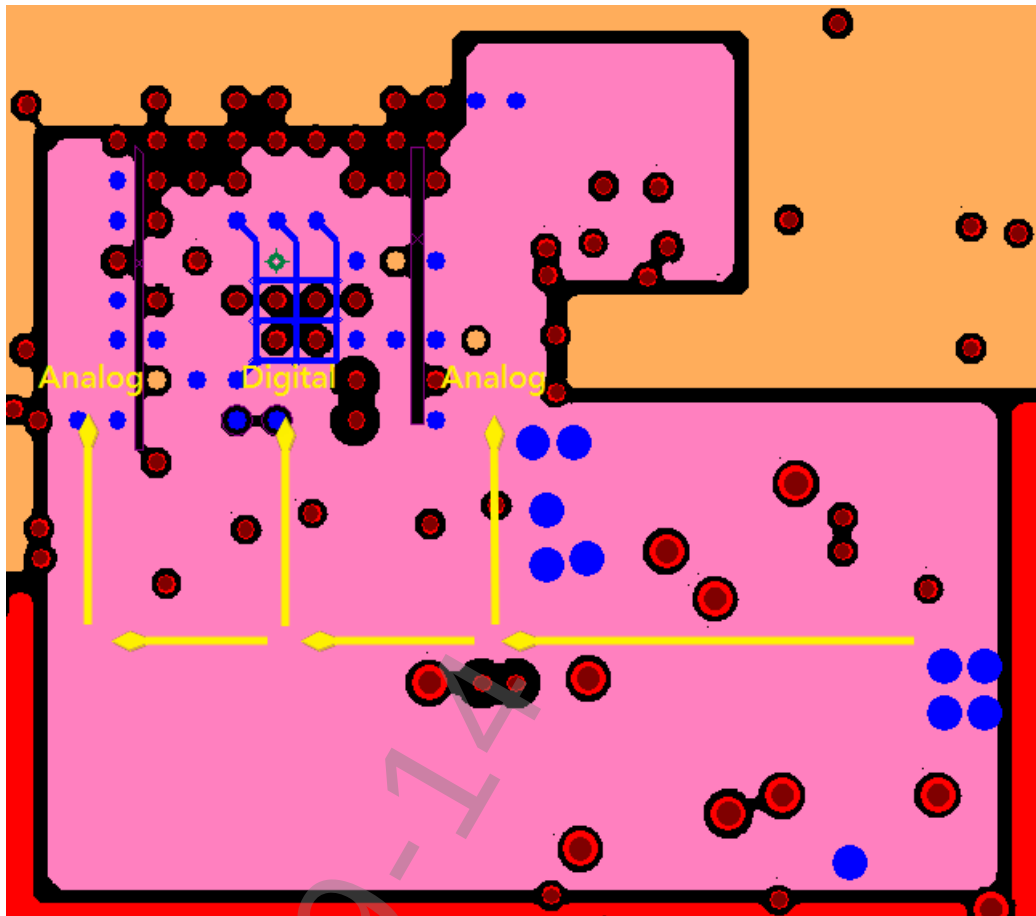


9.2 Layout rule

- Each power pin should be connected to a single 0.1uF capacitor as short as possible.
- The width of 5V $\geq$ 60 mil (suggest 80mil) .
- The width of VCCO , AVDDH $\geq$ 40 mil (suggest 50 mil) , VCCO is digital power , U_ AVDDH 、 SD_AVDD33 、 X_AVDDH are analog powers .
- The width of SD_AVDD18 $\geq$ 20 mil (suggest 30 mil) .
- The width of VCCCK , AVDDL $\geq$ 120 mil (suggest 150 mil) , VCCCK is digital power , P0_AVDDL 、 U_AVDDL 、 S_AVDDL 、 P1_AVDDL 、 SD_AVDDL 、 PLL_AVDDL are analog powers.
- Layout Topology :
 - Star topology: connect multiple powers to a central power source.
 - Daisy chaining topology : connect to digital power first , then connect to analog power .

9.3 Power Layout example

Layout example for Star topology



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